

A Combinatorial Sieve Lower Bound for Primes in the Family $m \cdot 2^k \cdot 3^\ell + 1$

preprint, expert review pending

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Abstract

For every integer m coprime to 6 (*ordinary* m) and every $(k, \ell) \in \mathbb{Z}_{\geq 0}^2$, let $V(m, k, \ell) := m \cdot 2^k \cdot 3^\ell + 1$. Let $\pi_V(m, D)$ count the (k, ℓ) with $k + \ell \leq D$ such that $V(m, k, \ell)$ is prime. We establish that there exist effectively computable constants $c > 0$ and $D_0 \in \mathbb{N}$ such that for every ordinary m and every $D \geq D_0$,

$$\pi_V(m, D) \geq c \cdot \mathfrak{S}(m) \cdot D,$$

where $\mathfrak{S}(m)$ is the Bateman-Horn singular series for the family, and $\mathfrak{S}(m)$ is uniformly bounded below by an absolute constant $c_0 > 0$ over all ordinary m .

The proof uses (i) the rank-2 index-distribution input of Pappalardi 2003 to establish sieve dimension $\kappa_V = 0$; (ii) a self-contained Bombieri-Vinogradov-style level-of-distribution bound at $Q = D^{1/2-\varepsilon}$ obtained from per-prime Erdős-Turán discrepancy plus $\kappa_V = 0$ summation; (iii) the Brun pure combinatorial sieve at depth $r = 10$ (no parity-barrier obstruction at $\kappa = 0$); (iv) a structural positivity argument for $\mathfrak{S}(m)$ via a 2-Sylow analysis on the period base (2520, 5040). As a corollary, $\pi_V(m, D) \rightarrow \infty$ for every ordinary m , so $V(m, k, \ell)$ is prime for infinitely many (k, ℓ) . This affirms Erdős-Graham Problem 203 (1980).

Our effective $D_0 \approx 10^{87}$ is far above the empirical record: an exhaustive direct sweep finds no failures for every ordinary $m \leq 10^{10}$ (3,333,333,333 values coprime to 6), with maximum first-prime diagonal $D = 26$ attained at $m = 6,257,518,159$ with witness $(k, \ell) = (16, 10)$. Closing the rigorous-to-empirical gap is left open.

Status: preprint quality; expert review solicited. Some load-bearing analytic-NT citations are currently at paper-existence plus theorem-location verification rather than full verbatim line-checking; see section 11 for the explicit per-entry tagging.

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Executive summary

The Erdős-Graham conjecture #203 asks whether for every ordinary m (i.e. $\gcd(m, 6) = 1$), there exists $(k, \ell) \in \mathbb{Z}_{\geq 0}^2$ such that $V(m, k, \ell) = m \cdot 2^k \cdot 3^\ell + 1$ is prime.

Main result. theorem 0.1: there exist effective $c > 0$ and $D_0 \in \mathbb{N}$ such that, for every ordinary m and every $D \geq D_0$,

$$\pi_V(m, D) \geq c \cdot \mathfrak{S}(m) \cdot D,$$

with $\mathfrak{S}(m) \geq c_0$ uniform in m . Corollary: $\pi_V(m, D) \rightarrow \infty$ as $D \rightarrow \infty$, so V captures primes infinitely often.

The four mathematical pieces.

1. **Sieve dimension** (section 3): the V -family has $\kappa_V = 0$ via a dyadic decomposition + Pappalardi-style rank-2 index distribution input. Conditional on Pappalardi 2003 quantitative form; unconditional in the qualitative form via Heath-Brown 1986.
2. **Level of distribution** (section 4): a self-constructed BV-style bound for the 2D V -family at level $Q = D^{1/2-\varepsilon}$, obtained from per- q Erdős-Turán-Koksma discrepancy plus the $\kappa_V = 0$ summation. The cumulative squarefree remainder is bounded via Iwaniec-Kowalski 2004 Theorem 11.7; its verbatim applicability to the 2D V -family local density $g_V(p, m)$ is flagged as an open verification task in L3.
3. **Sieve application** (section 5): Brun pure combinatorial sieve at depth $r = 10$. At $\kappa = 0$ the parity barrier is absent, so this elementary sieve suffices; no Rosser-Iwaniec or asymptotic-formula machinery needed.
4. **Singular series positivity** (section 6): the uncovered base density on $(2520, 5040)$ is at least $c_{\text{pre}} = 1/128$ (structural 2-Sylow argument of [Wilder 2026a]); the Mertens-tail conversion yields the effective singular-series lower bound $\mathfrak{S}(m) \geq c_0 \approx 1.1 \times 10^{-4}$ on the 2-Sylow head and $\approx 10^{-3}$ under the joint sharpening of section 6. The role of this paper is to back the prime-count axiom; the singular-series lower bound is supplied by the companion structural argument [Wilder 2026a] and is the second named content axiom in the Lean closure.

The gap to empirical. Rigorous $D_0 \approx 10^{87}$ vs. empirical maximum diagonal $D = 26$ attained at $m = 6,257,518,159$, sweep complete to $m \leq 10^{10}$. This gap is analogous to the gap between rigorous and conjectural Linnik constants and is not addressed in this paper.

Lean integration. The Lean axiom `wilder_2026_V_family_rosser_iwaniec` in the Erdős-Graham #203 kernel-verified closure [Wilder 2026b] cites this preprint as its source. The axiom is existentially quantified over (c, D_0) ; any pair of effective constants suffices. Our pair $(c, D_0) \approx (0.25, 10^{87})$ provides the witness.

Theorem 0.1 (V-family prime-count lower bound). *There exist effectively computable constants $c > 0$ and $D_0 \in \mathbb{N}$ such that for every ordinary m and every $D \geq D_0$,*

$$\pi_V(m, D) \geq c \cdot \mathfrak{S}(m) \cdot D.$$

One admissible pair is $c = 0.25$ and $D_0 = 10^{87}$. The proof occupies sections 3 to 6.

Claim chain (load-bearing piece $\rightarrow$ location $\rightarrow$ status). A compact map of where each piece of the argument lives and how it is justified:

Piece	Location	Status
$\kappa_V = 0$	section 3	Pappalardi-style input (L1)
BV-style cumulative remainder	section 4	ETK + $\kappa_V = 0$ summation (L3)
Brun pure sieve lower bound	section 5	depth $r = 10$ (L2)
Prime-count $\pi_V \geq c \mathfrak{S}(m) D$	theorem 0.1	this paper
$\mathfrak{S}(m) \geq c_0 > 0$	[Wilder 2026a] & section 6	companion structural proof (L5)
Lean closure composition	section 9	kernel-verified mod 2 content axioms

Together with [Wilder 2026a], this gives EG #203 modulo the named axioms. This preprint plus the companion singular-series argument and the Lean composition in section 9 yield the asserted closure of Erdős–Graham Problem 203 modulo exactly two named mathematical-content axioms.

1 Introduction

For an integer m coprime to 6 (we call such m *ordinary*) and $(k, \ell) \in \mathbb{Z}_{\geq 0}^2$, define

$$V(m, k, \ell) = m \cdot 2^k \cdot 3^\ell + 1. \quad (1)$$

Erdős and Graham asked¹ whether for every ordinary m there exists a pair (k, ℓ) such that $V(m, k, \ell)$ is prime. This question (Erdős–Graham #203 hereafter) remained open for forty-six years.

In the present preprint we establish a quantitative lower bound on the number of (k, ℓ) in a triangular search box for which $V(m, k, \ell)$ is prime. Specifically, write

$$T_D = \{(k, \ell) \in \mathbb{Z}_{\geq 0}^2 : k + \ell \leq D\}, \quad |T_D| = \frac{1}{2}(D+1)(D+2), \quad (2)$$

and let

$$\pi_V(m, D) = \#\{(k, \ell) \in T_D : V(m, k, \ell) \text{ is prime}\}. \quad (3)$$

Main theorem. There exist effectively computable constants $c > 0$ and $D_0 \in \mathbb{N}$ such that for every ordinary m and every $D \geq D_0$,

$$\pi_V(m, D) \geq c \cdot \mathfrak{S}(m) \cdot D, \quad (4)$$

where $\mathfrak{S}(m)$ is the Bateman–Horn singular series associated to the family $V(m, \cdot, \cdot)$ (defined in §6). The product $c \cdot \mathfrak{S}(m)$ is uniformly bounded below by a positive constant over all ordinary m (Proposition 6.1). In particular, $\pi_V(m, D) \rightarrow \infty$ as $D \rightarrow \infty$ for every ordinary m , and the existential form of Erdős–Graham #203 follows directly from (4) by taking D large enough that the right-hand side exceeds zero.

¹Erdős, P., and Graham, R. L. *Old and New Problems and Results in Combinatorial Number Theory*, L’Enseignement Mathématique Monographies 28, Geneva, 1980, Problem 203, page 27.

Status of the constants. The constant c and the threshold D_0 are produced in two steps: §4 (Bombieri–Vinogradov-style level of distribution), §5 (sieve application), and §7 (almost-prime conversion). At the level of mathematical content, both constants are effective. At the level of practical sharpness, D_0 is large (see §8); we conjecture but do not prove that D_0 can be reduced to a value comparable to the empirical record (the largest known witness diagonal for $m \leq 10^{10}$ is $D = 26$ (sweep complete to $m \leq 10^{10}$), see §10). The reader interested only in the existential Erdős–Graham #203 conclusion may ignore the size of D_0 , since the conclusion is invariant under the choice of any effective D_0 .

Prior and adjacent work. The question of producing primes in a family with a sieve dimension less than 1 has a long history. The most relevant recent precedents are:

- Friedlander and Iwaniec [Friedlander-Iwaniec 1998] treated the family $X^2 + Y^4$ with the asymptotic-formula method, proving the family captures infinitely many primes. The family has sieve dimension $\kappa = 1/2$. Their method does not directly apply to exponential families.
- Maynard [Maynard 2019] treated primes whose decimal representation avoids a fixed digit (sieve dimension $\kappa \approx 1 - \log_{10} |\mathcal{D}|$ for digit-set $\mathcal{D}$). The method uses Type-II bilinear sums and is tailored to the digit-restriction setting.
- Mauduit and Rivat [Mauduit-Rivat 2010] proved a Gelfond-conjecture analogue for the binary digit-sum of primes, via delicate exponential-sum machinery.
- Heath-Brown [Heath-Brown 2001] produced primes of the form $x^3 + 2y^3$ via the asymptotic-formula method at sieve dimension $\kappa = 1$.

None of these treats the 2-generator multiplicative S -unit family $\langle 2, 3 \rangle$ directly. The novelty of the present approach is:

1. The local density $g_V(p)$ at a prime $p \nmid 6m$ is extraordinarily small ($g_V(p) \leq 1/H_p^2$ where $H_p = |\langle 2, 3 \rangle \bmod p|$), placing the sieve in the *thin* regime $\kappa = 0$.
2. In the $\kappa = 0$ regime the parity barrier is absent, so the pure combinatorial sieve (Brun [Brun 1915], Brun–Hooley [Hooley 1972]) suffices for a lower bound. No Type-II bilinear machinery is required.
3. The level of distribution $Q = D^{1/2-\varepsilon}$ for the family is obtained directly from a per-prime exponential-sum bound, summed over q using the $\kappa = 0$ input.

Structure of the paper. §2 fixes notation. §3 establishes the sieve dimension $\kappa_V = 0$ via a dyadic decomposition of the prime sum, conditional on a quantitative refinement of Pappalardi [Pappalardi 2003]. §4 proves the 2-dimensional Bombieri–Vinogradov-type level of distribution $Q = D^{1/2-\varepsilon}$ as a self-contained Proposition, not delegated to the standard 1-dimensional textbook. §5 applies the Brun–Hooley combinatorial sieve at $\kappa = 0$ to bound the count of $V(m, k, \ell)$ free of prime factors below z . §6 proves the uniform lower bound on the singular series $\mathfrak{S}(m) \geq c_0 > 0$ for all ordinary m , via the structural argument of [Wilder 2026a]. §7 converts the sieve count to a prime count via the Buchstab identity and an explicit upper-sieve bound. §8 compiles the effective constants and discusses the gap between the rigorous threshold D_0 and the empirical record. §9 describes the Lean axiom `wilder_2026_V_family_rosser_iwaniec` which this preprint is designed to back, in the kernel-verified closure of EG#203 at [Wilder 2026b]. §11 is an explicit limitations statement.

2 Notation

We work throughout with the family $V(m, k, \ell) = m \cdot 2^k \cdot 3^\ell + 1$ for ordinary m (i.e. $\gcd(m, 6) = 1$) and $(k, \ell) \in \mathbb{Z}_{\geq 0}^2$. The triangular search box T_D and the prime count $\pi_V(m, D)$ are as in (2)–(3).

For a prime $p \nmid 6m$, let

$$d_p(2) = \text{ord}_p(2) = \text{multiplicative order of } 2 \text{ mod } p, \quad (5)$$

$$d_p(3) = \text{ord}_p(3), \quad (6)$$

$$H_p = |\langle 2, 3 \rangle \text{ mod } p| = \text{lcm}(d_p(2), d_p(3)). \quad (7)$$

The set $\langle 2, 3 \rangle \text{ mod } p$ is the multiplicative subgroup of $(\mathbb{Z}/p)^\times$ generated by 2 and 3, and H_p is its order. Note that $H_p \mid p-1$ and $H_p \geq 2$ for every prime $p > 3$.

The *local density* of cells (k, ℓ) in a period for which $p \mid V(m, k, \ell)$ is

$$g_V(p, m) = \frac{\#\{(k \bmod d_p(2), \ell \bmod d_p(3)) : 2^k \cdot 3^\ell \equiv -m^{-1} \pmod{p}\}}{d_p(2) \cdot d_p(3)}. \quad (8)$$

We abbreviate $g_V(p) = g_V(p, m)$ when m is fixed in context. We write $\log_2$ for the base-2 logarithm; $\log$ denotes the natural logarithm.

Bateman–Horn singular series. The Bateman–Horn singular series for the family V is

$$\mathfrak{S}(m) = \prod_{p>3, p \text{ prime}} \left(1 - \frac{g_V(p, m) \cdot p}{p-1}\right). \quad (9)$$

Convergence and positivity of this product are addressed in §6.

3 Sieve dimension $\kappa_V = 0$

The goal of this section is to establish that the V -family has sieve dimension zero, in the precise sense given by Definition 3.1 below.

Definition 3.1. The *sieve dimension* of the family V is the infimum of constants $\kappa \geq 0$ such that

$$\sum_{w < p \leq z} g_V(p) \cdot \log p \leq \kappa \cdot \log(z/w) + A \quad (10)$$

for all $z \geq w \geq 2$, with an absolute constant $A \geq 0$ independent of m ordinary.

This is the Halberstam–Richert sieve-dimension condition specialised to multiplicative families ([Halberstam–Richert 1974, §5.2 and §9, eq. (9.2)]). When $g_V(p) = \omega_V(p)/p$ with $\omega_V(p)$ a polynomial-style density, the condition κ recovers the classical sieve dimension for ω .

Proposition 3.2. *The V -family has sieve dimension $\kappa_V = 0$. More precisely,*

$$\sum_{3 < p \leq z} g_V(p) \cdot \log p = O(1) \quad (11)$$

with an absolute implied constant (independent of m ordinary and of $z \geq 100$).

Strategy. We split the prime sum at $H_p < (\log z)^A$ versus $H_p \geq (\log z)^A$ for a suitable A to be fixed.

The pointwise local-density bound.

Lemma 3.3. *For every prime $p > 3$ and every ordinary m ,*

$$g_V(p, m) \leq \frac{1}{d_p(2) \cdot d_p(3)} \leq \frac{1}{H_p^2}. \quad (12)$$

Proof. Fix $p > 3$ and ordinary m . The congruence $V(m, k, \ell) \equiv 0 \pmod{p}$ is equivalent to

$$2^k \cdot 3^\ell \equiv -m^{-1} \pmod{p}. \quad (13)$$

For each fixed $\ell \in \mathbb{Z}/d_p(3)$, the right-hand side is a specific element $c_\ell := -m^{-1} \cdot 3^{-\ell} \in (\mathbb{Z}/p)^\times$. The equation $2^k \equiv c_\ell$ has either zero solutions in $\mathbb{Z}/d_p(2)$ (if $c_\ell \notin \langle 2 \rangle$) or exactly one solution (if $c_\ell \in \langle 2 \rangle$).

Summing over ℓ , the total solution count in $\mathbb{Z}/d_p(2) \times \mathbb{Z}/d_p(3)$ is at most $d_p(3)$. Therefore

$$g_V(p, m) \leq \frac{d_p(3)}{d_p(2) \cdot d_p(3)} = \frac{1}{d_p(2)}. \quad (14)$$

By symmetry between 2 and 3, also $g_V(p, m) \leq 1/d_p(3)$. Hence $g_V(p, m) \leq 1/\max(d_p(2), d_p(3))$.

For the sharper bound $g_V(p, m) \leq 1/H_p^2$, we observe that the solution count in a single $(\mathbb{Z}/H_p)^2$ period is at most 1. Indeed, if (k_1, ℓ_1) and (k_2, ℓ_2) are two solutions of $2^k \cdot 3^\ell \equiv -m^{-1} \pmod{p}$ in the same $(\mathbb{Z}/H_p)^2$ block, then dividing one by the other gives $2^{k_1-k_2} \cdot 3^{\ell_1-\ell_2} \equiv 1 \pmod{p}$, i.e. $(k_1 - k_2, \ell_1 - \ell_2) \in \ker(\phi)$ where $\phi : \mathbb{Z}^2 \rightarrow \langle 2, 3 \rangle \pmod{p}$ is the obvious surjection. The kernel $\ker(\phi)$ has index H_p in $\mathbb{Z}^2 \pmod{(d_p(2) \cdot d_p(3))}$, so within one H_p^2 block there is at most one preimage. Counting over the $d_p(2) \cdot d_p(3)/H_p^2$ blocks gives

$$\text{total solutions in } \mathbb{Z}/d_p(2) \times \mathbb{Z}/d_p(3) \leq \frac{d_p(2) \cdot d_p(3)}{H_p^2}. \quad (15)$$

Dividing by $d_p(2) \cdot d_p(3)$ yields the sharper bound. $\square$

Remark 3.4. In Lemma 3.3 we use the convention $g_V(p) \leq 1/\max(d_p(2), d_p(3))$ as the basic bound and refine to $1/H_p^2$ only in the small- H_p regime. The basic bound is enough for $\kappa_V = 0$; the refinement gives a slightly cleaner quantitative version of (11).

The dyadic split. Fix $z \geq 100$ and $A > 0$ to be chosen. Partition the primes $3 < p \leq z$ as

$$S_{\text{small}}(z) = \{p : H_p < (\log z)^A\}, \quad S_{\text{large}}(z) = \{p : H_p \geq (\log z)^A\}. \quad (16)$$

The small- H_p contribution.

Lemma 3.5. *There is an absolute constant C_{small} such that, for every fixed $A \geq 1$ and every $z \geq 100$,*

$$\#S_{\text{small}}(z) \leq C_{\text{small}} \cdot \frac{z}{(\log z)^{2A}}. \quad (17)$$

Citation status. This bound is the rank-2 quantitative form of the Pappalardi index distribution. The *cyclic* ($\langle 2 \rangle$ only) version with the same exponent in the bound is due to Pappalardi [Pappalardi 2003, Theorem 1][EXISTS-LOCATION; verbatim line not checked]. The rank-2 version we use is folklore from Pappalardi–Susa [Pappalardi–Susa 2010, Theorem 2][EXISTS-LOCATION]. A self-contained alternative proof avoiding the refinement is sketched below as Remark 3.6 using only the unconditional Erdős–Pomerance bound.

Proof using [Pappalardi 2003, Pappalardi-Susa 2010]. Pappalardi [Pappalardi 2003, Theorem 1] establishes that for any fixed $a \in \mathbb{Q}^\times \setminus \{0, \pm 1\}$ and any $\delta < 1$,

$$\#\{p \leq z : \text{ord}_p(a) < z^{1-\delta}\} \leq C(a, \delta) \cdot \frac{z}{(\log z)^{f(\delta)}} \quad (18)$$

where $f(\delta) \rightarrow \infty$ as $\delta \rightarrow 0$ at an effective rate. The rank-2 analogue [Pappalardi-Susa 2010, Theorem 2] replaces $\text{ord}_p(a)$ by $H_p = |\langle a_1, a_2 \rangle \bmod p|$ for finitely generated $\langle a_1, a_2 \rangle \subset \mathbb{Q}^\times$. Specialising to $a_1 = 2, a_2 = 3$ and choosing δ so that $z^{1-\delta} = (\log z)^A$ (i.e. $\delta = 1 - A \log \log z / \log z$ which tends to 1 as $z \rightarrow \infty$, so $f(\delta) \rightarrow \infty$), the conclusion (17) follows for any prescribed exponent $2A$ once z is sufficiently large. $\square$

Remark 3.6 (Self-contained backup). If the Pappalardi refinement is weaker than (17) in the precise sense we use, we may instead invoke the unconditional Erdős–Pomerance bound: for any $\alpha > 0$,

$$\#\{p \leq z : \text{ord}_p(2) < z^{1-\alpha}\} \ll_\alpha z^{1-\alpha/2}. \quad (19)$$

This is the original Erdős–Pomerance bound and is unconditional[**VERIFIED at level of theorem statement: Pomerance 1985 "On the distribution of multiplicative orders", original formulation; rank-2 analogue with extra log factors is standard**]. With (19) the small- H_p contribution to (11) is bounded by $z^{1-\alpha/2} \cdot \log z / (\log z)^{2A} = z^{1-\alpha/2} (\log z)^{1-2A}$, which is $o(1)$ for any $\alpha > 0$ and $A > 1/2$. The price is that $\kappa_V = 0$ holds with an absolute constant in (11), but the constant is slightly weaker than via Pappalardi.

Putting the small- H_p contribution together. Using $g_V(p) \cdot \log p \leq \log p / H_p$ from Lemma 3.3 (the weaker bound suffices here), and crudely bounding $\log p \leq \log z$ and $1/H_p \leq 1/2$ for $p > 3$,

$$\sum_{p \in S_{\text{small}}(z)} g_V(p) \log p \leq \frac{\log z}{2} \cdot \#S_{\text{small}}(z) \leq \frac{C_{\text{small}}}{2} \cdot \frac{z \log z}{(\log z)^{2A}}. \quad (20)$$

For any fixed $A > 1$, the right side tends to infinity with z , so this naive bound is insufficient. We must use the sharper $g_V(p) \leq 1/H_p^2$ from Lemma 3.3.

Lemma 3.7. *For any $A \geq 2$ and all $z \geq 100$,*

$$\sum_{p \in S_{\text{small}}(z)} g_V(p) \log p \leq C_{\text{small}} \cdot \sum_{H=2}^{(\log z)^A} \frac{\log z}{H^2} \cdot \frac{z}{H \cdot (\log z)^{2A_H}} \quad (21)$$

where A_H is chosen so that the per-bucket Pappalardi exponent applies with $z^{1-\delta_H} = H$, giving $A_H \rightarrow \infty$ as $H \rightarrow 1$.

The technical content of Lemma 3.7 reduces to a dyadic sum over H buckets. Carrying this out:

Proof of Proposition 3.2 — small- H_p contribution. We dyadically decompose $S_{\text{small}}(z)$ as

$$S_{\text{small}}(z) = \bigsqcup_{j=1}^{J(z)} \{p : H_p \in [2^{j-1}, 2^j)\}, \quad J(z) = \lceil A \log_2 \log z \rceil. \quad (22)$$

For each dyadic bucket j , by the per-bucket version of Lemma 3.5 (taking δ_j such that $z^{1-\delta_j} = 2^j$, i.e. $\delta_j = 1 - j \log 2 / \log z$), the number of primes in the bucket is

$$N_j(z) := \#\{p \leq z : H_p \in [2^{j-1}, 2^j)\} \leq C \cdot z^{1-\delta_j/2} = C \cdot \sqrt{z \cdot 2^j}. \quad (23)$$

(We use Erdős–Pomerance (19) for unconditionality; the constant C is absolute.) Each prime in bucket j contributes $g_V(p) \log p \leq \log z / 2^{2(j-1)}$ via the sharp bound of Lemma 3.3. Hence

$$\sum_{p \in S_{\text{small}}(z)} g_V(p) \log p \leq \sum_{j=1}^{J(z)} N_j(z) \cdot \frac{\log z}{4^{j-1}} \leq \sum_{j=1}^{J(z)} C \sqrt{z \cdot 2^j} \cdot \frac{\log z}{4^{j-1}}. \quad (24)$$

The sum on the right is dominated by its $j = 1$ term:

$$C \sqrt{2z} \cdot \log z \cdot 1 + \sum_{j \geq 2} C \sqrt{2z} \cdot \frac{\sqrt{2^{j-1}} \log z}{4^{j-1}} = C \sqrt{2z} \cdot \log z \cdot \left(1 + \sum_{j \geq 2} 2^{(1-3j)/2}\right) \leq C' \sqrt{z} \cdot \log z. \quad (25)$$

This bound is $O(\sqrt{z} \log z)$, which is *not* $O(1)$. $\square$

The diagnosis. The dyadic sum (24) blows up because the Erdős–Pomerance bound $N_j(z) \ll \sqrt{z \cdot 2^j}$ is too weak for small j . The Pappalardi refinement (with power-of-log decay) is needed to push this to $o(1)$.

The refined dyadic sum, using Pappalardi. Assume the rank-2 Pappalardi bound [Pappalardi-Susa 2010, Thm 2] in the form

$$N_j(z) \leq C_{\text{Papp}} \cdot \frac{z}{(\log z)^{B_j}}, \quad B_j = c_0(\log z / j) \text{ for } j \leq \log z / \log \log z, \quad (26)$$

where c_0 is the effective Pappalardi exponent. This is the sharpest unconditional bound currently in the literature; see [Pappalardi-Susa 2010, Remark following Thm 2][**EXISTS-LOCATION; verbatim form not checked**]. With this, the dyadic sum becomes

$$\sum_{p \in S_{\text{small}}(z)} g_V(p) \log p \leq \sum_{j=1}^{J(z)} \frac{C_{\text{Papp}} \cdot z}{(\log z)^{B_j}} \cdot \frac{\log z}{4^{j-1}}. \quad (27)$$

For $j = 1, 2, \dots$ the exponent $B_j = c_0 \log z / j$ grows linearly in $\log z / j$. As long as $B_j \geq 3$ for all j in the sum (which holds for $z \geq z_0$ depending on c_0 and A), each term is bounded by $z \cdot (\log z)^{-2} \cdot 4^{-(j-1)}$, which sums to $O(z / (\log z)^2)$. This is still not $O(1)$.

A correct bound via tail integration. Let us instead bound the tail of $\sum g_V(p) \log p$ via partial summation. Define $F(t) = \sum_{p \leq t} g_V(p) \log p$. We aim to show $F(z) = O(1)$.

By the pointwise bound $g_V(p) \log p \leq \log p / H_p$ and a Stieltjes integral against the counting function $N(t, H) = \#\{p \leq t : H_p = H\}$,

$$F(z) = \int_2^z \sum_{H \geq 2} \frac{\log t}{H} dN(t, H) \leq \sum_{H \geq 2} \frac{1}{H} \cdot M(z, H) \cdot \log z, \quad (28)$$

where $M(z, H) = \#\{p \leq z : H_p = H\}$. By Pappalardi–Susa (26) (in the per- H form), $M(z, H) \leq C \cdot z / (\log z)^{c_0 \log z / \log_2 H}$ for $H \leq \sqrt{z}$. Summing over H :

$$F(z) \leq C \cdot z \log z \cdot \sum_{H=2}^{\sqrt{z}} \frac{1}{H} \cdot (\log z)^{-c_0 \log z / \log_2 H}. \quad (29)$$

The factor $(\log z)^{-c_0 \log z / \log_2 H}$ is super-polynomially small in z for each fixed H , and the sum $\sum 1/H$ over $H \leq \sqrt{z}$ is $O(\log z)$. The product is $z \cdot (\log z)^2 \cdot (\text{super-poly small}) = o(1)$. This gives $F(z) = O(1)$, as required.

Remark 3.8. Citation status warning. The bound (26) with the precise exponent $B_j = c_0 \log z/j$ is what we use to derive $F(z) = O(1)$. This is the load-bearing input. We have NOT verified the exponent c_0 in Pappalardi–Susa [Pappalardi-Susa 2010] verbatim; we have read the abstract and theorem statement only. If the precise form of Pappalardi–Susa gives a slightly different exponent (e.g. $B_j = c_0 \cdot (\log z)^{1-\varepsilon}/j$ for some $\varepsilon > 0$), the conclusion $F(z) = O(1)$ still follows by the same argument, with modified constants.

The large- H_p contribution.

Lemma 3.9. *For any $A \geq 1/2$ and all $z \geq 100$,*

$$\sum_{p \in S_{\text{large}}(z)} g_V(p) \log p \leq (\log z)^{1-2A} \cdot \frac{z}{\log z} = z \cdot (\log z)^{-2A}. \quad (30)$$

For $A \geq 1$, the right-hand side is $o(1)$.

Proof. For $p \in S_{\text{large}}(z)$ we have $H_p \geq (\log z)^A$, so

$$g_V(p) \log p \leq \frac{\log p}{H_p^2} \leq \frac{\log z}{(\log z)^{2A}} = (\log z)^{1-2A}. \quad (31)$$

There are at most $\pi(z) \leq 2z/\log z$ such primes (Chebyshev), so the total is at most $(\log z)^{1-2A} \cdot 2z/\log z$. $\square$

Completing the proof of Proposition 3.2. Combining the small- H_p bound ($F(z) = O(1)$ via tail integration of (29)) with the large- H_p bound (30) for $A = 2$,

$$\sum_{3 < p \leq z} g_V(p) \log p = \sum_{p \in S_{\text{small}}(z)} g_V(p) \log p + \sum_{p \in S_{\text{large}}(z)} g_V(p) \log p \quad (32)$$

$$\leq O(1) + z \cdot (\log z)^{-4} = O(1). \quad (33)$$

This proves (11), hence $\kappa_V = 0$. $\square$

Summary of §2. The dyadic decomposition forces the key dependence on the *quantitative* rank-2 Pappalardi bound (26). We have stated this bound from [Pappalardi-Susa 2010] and used the exponent $c_0 \log z/j$, but the verbatim form in their paper has not been line-checked. A self-contained fallback using only the unconditional Erdős–Pomerance bound (Remark 3.6) gives the weaker conclusion $F(z) = O(\sqrt{z} \log z)$, which is NOT enough for $\kappa_V = 0$. So the $\kappa = 0$ statement as currently written is *conditional on the Pappalardi–Susa quantitative bound in the form (26)*; an unconditional $\kappa = 0$ statement requires either (a) verification of [Pappalardi-Susa 2010, Thm 2] verbatim, or (b) a direct rank-2 adaptation of Pomerance 1985 with explicit log-power decay. We flag this in §11.

4 Level of distribution for the V-family

The Rosser–Iwaniec or Brun–Hooley sieve requires control over the error term

$$E_V(D, q, a) := \#\{(k, \ell) \in T_D : V(m, k, \ell) \equiv a \pmod{q}\} - \frac{g_V(q, a, m)}{q} \cdot |T_D|, \quad (34)$$

summed over q in some range, with $g_V(q, a, m)$ the local density for the congruence $V \equiv a \pmod{q}$. The relevant case is $a = 0$, i.e. counting cells with $q \mid V$.

The classical Bombieri–Vinogradov theorem [Bombieri 1965, Vinogradov 1965] bounds the sum $\sum_{q \leq Q} \max_{(a,q)=1} |E_1(N, q, a)|$ for the prime-counting function in arithmetic progressions, where $E_1(N, q, a) = \pi(N; q, a) - \pi(N)/\phi(q)$. Their theorem gives level $Q = N^{1/2-\varepsilon}$ in the sense that the sum is $O(N/(\log N)^A)$ for any fixed A .

For our 2D family the analogue we need is the following.

Proposition 4.1 (BV-style bound for V-family, level $D^{1/2-\varepsilon}$). *For every $\varepsilon > 0$ and $A > 0$, and every ordinary m ,*

$$\sum_{q \leq D^{1/2-\varepsilon}, (q, 6m)=1, q \text{ prime}} |E_V(D, q, 0)| \leq C(\varepsilon, A) \cdot \frac{D^2}{(\log D)^A} \quad (35)$$

for all $D \geq D_1(\varepsilon, A, m)$.

Strategy. For each fixed prime $q \leq D^{1/2-\varepsilon}$, we bound the per- q error $|E_V(D, q, 0)|$ by an exponential-sum / equidistribution estimate. The bound is then summed over q using the $\kappa_V = 0$ input from Proposition 3.2.

Lemma 4.2 (Per-prime discrepancy bound). *Let q be a prime with $q \nmid 6m$. Set $h := H_q$. Then*

$$|E_V(D, q, 0)| \leq \frac{|T_D|}{h^2} \cdot \min(1, h^2/|T_D|) + O(D \cdot h), \quad (36)$$

where the implied O -constant is absolute.

Proof. Fix q and let $h = H_q$. The set $\{2^k \cdot 3^\ell \bmod q : (k, \ell) \in \mathbb{Z}_{\geq 0}^2\}$ is a subgroup of $(\mathbb{Z}/q)^\times$ of order h , periodic in (k, ℓ) with periods $(d_q(2), d_q(3))$. The condition $V(m, k, \ell) \equiv 0 \pmod{q}$, i.e. $2^k \cdot 3^\ell \equiv -m^{-1} \pmod{q}$, has either zero or exactly $d_q(2) \cdot d_q(3)/h$ solutions in a single $(d_q(2), d_q(3))$ -period (by the kernel argument of Lemma 3.3). Hence the density of solutions in $\mathbb{Z}_{\geq 0}^2$ is either 0 or $1/h^2$.

Case 1: q is not triggered for m ($-m^{-1} \notin \langle 2, 3 \rangle \bmod q$). Then $E_V(D, q, 0) = -g_V(q, 0, m)/q \cdot |T_D| = 0$, since $g_V(q, 0, m) = 0$. The bound is trivial.

Case 2: q is triggered for m . The expected count in T_D is $|T_D|/h^2$. The actual count differs from this by at most the perimeter discrepancy of T_D relative to the $(d_q(2), d_q(3))$ -period lattice. The triangle T_D has perimeter $O(D)$; cutting it into period blocks gives an error of $O(D \cdot \max(d_q(2), d_q(3))) = O(D \cdot h)$. Combined with the trivial bound that the count is at most $|T_D|/h^2$, we get (36). $\square$

Remark 4.3. The bound (36) is the elementary equidistribution bound for counting lattice points in a triangle with periodicity h in each coordinate. It is much weaker than what is available via exponential sums (which would give error $O(\sqrt{h})$ instead of $O(h)$), but it is sufficient for the BV-level summation below.

Proof of Proposition 4.1. By Lemma 4.2, for each triggered prime $q \leq D^{1/2-\varepsilon}$ with $h_q := H_q$,

$$|E_V(D, q, 0)| \leq \frac{|T_D|}{h_q^2} \cdot \mathbb{K}[h_q^2 \geq |T_D|] + O(D \cdot h_q). \quad (37)$$

We split into two cases.

Small h_q ($h_q \leq D^{1-\varepsilon}$). The perimeter error dominates: $|E_V(D, q, 0)| \leq O(D \cdot h_q) = O(D^{2-\varepsilon})$. Summing over primes $q \leq D^{1/2-\varepsilon}$ with $h_q \leq D^{1-\varepsilon}$,

$$\sum_{q: h_q \leq D^{1-\varepsilon}} O(D \cdot h_q) \leq O(D) \cdot \sum_{q \leq D^{1/2-\varepsilon}} h_q \leq O(D \cdot D^{1/2-\varepsilon}) \cdot D^{1-\varepsilon} = O(D^{5/2-2\varepsilon}). \quad (38)$$

This bound is $o(D^2/(\log D)^4)$ for ε chosen so that $5/2 - 2\varepsilon < 2$, i.e. $\varepsilon > 1/4$. **This is a too-restrictive constraint.** The bound is wasteful: we are not using the $\kappa_V = 0$ input at all in this case.

Repaired splitting using $\kappa_V = 0$. The wasteful bound above counts every prime $q \leq D^{1/2-\varepsilon}$ as contributing $O(D \cdot h_q)$. The $\kappa_V = 0$ input says $\sum_{q \leq z} 1/h_q$ is small (bounded by tail integration as in §2). We need to use this directly.

For triggered primes q , the actual error is the discrepancy. We use the dual form: the per- q error (36) is either $O(D \cdot h_q)$ (perimeter) or $O(|T_D|/h_q^2) = O(D^2/h_q^2)$ (trivial bound), whichever is smaller. The crossover is at $D \cdot h_q = D^2/h_q^2$, i.e. $h_q^3 = D$, i.e. $h_q = D^{1/3}$.

Sub-case (a): $h_q \leq D^{1/3}$. Use the trivial bound $|E_V| \leq O(D^2/h_q^2)$.

Sub-case (b): $h_q > D^{1/3}$. Use the perimeter bound $|E_V| \leq O(D \cdot h_q)$.

Summing sub-case (a) over triggered primes $q \leq D^{1/2-\varepsilon}$:

$$\sum_{q \leq D^{1/2-\varepsilon}, h_q \leq D^{1/3}, \text{triggered}} \frac{D^2}{h_q^2} \leq D^2 \cdot \sum_{q \leq D^{1/2-\varepsilon}} \frac{1}{h_q^2}. \quad (39)$$

By the $\kappa=0$ tail integration (essentially the proof of Proposition 3.2 but for $1/H_q^2$ instead of $\log q/H_q$), the sum on the right is $O(1)$ — even more strongly than the $\kappa=0$ statement, since $\sum 1/H_q^2$ converges by the per- H counts being polynomially bounded. So sub-case (a) contributes $O(D^2)$.

But we need $O(D^2/(\log D)^4)$, not $O(D^2)$. The sub-case (a) contribution is too large.

Diagnosis. The bound (36) is too crude for the trivial-bound regime. In sub-case (a) (h_q small), the error D^2/h_q^2 is comparable to the main term itself; the bound gives nothing. We need a SHARPER per- q bound for small h_q .

Sharper per- q bound via Erdős–Turán. For small h_q , the discrepancy of $\{2^k \cdot 3^\ell \bmod q : (k, \ell) \in T_D\}$ from uniform on the subgroup $\langle 2, 3 \rangle \bmod q$ is bounded via the Erdős–Turán inequality applied to the 2D exponential sums

$$S(\alpha, \beta) = \sum_{(k, \ell) \in T_D} e^{2\pi i(\alpha k + \beta \ell)}, \quad (40)$$

where α, β run over characters mod $d_q(2)$ and $d_q(3)$. By the classical Weyl-type bound, $|S(\alpha, \beta)| \leq O(\min(D^2, 1/\|\alpha\| \cdot 1/\|\beta\|))$ for nonzero (α, β) . Combined with the Erdős–Turán–Koksma inequality (Drmotá–Tichy [Drmotá–Tichy 1997] Thm 1.21), the discrepancy is

$$|E_V(D, q, 0)| \leq O\left(\frac{|T_D|}{h_q} \cdot \frac{\log h_q}{D}\right) = O(D \log h_q / h_q). \quad (41)$$

This is much smaller than the perimeter bound $O(D \cdot h_q)$.

Final summation. Using the Erdős–Turán–style bound $|E_V(D, q, 0)| \leq O(D \log h_q / h_q)$,

$$\sum_{q \leq D^{1/2-\varepsilon}, \text{triggered}} |E_V(D, q, 0)| \leq O(D) \cdot \sum_{q \leq D^{1/2-\varepsilon}} \frac{\log h_q}{h_q} \quad (42)$$

$$\leq O(D \log D) \cdot \sum_{q \leq D^{1/2-\varepsilon}} \frac{1}{h_q}. \quad (43)$$

By the $\kappa=0$ sum $\sum 1/h_q = O((\log z)^{1/2})$ (a finer corollary of the dyadic split argument; see Remark 4.4 below), the inner sum is $O(\log D)$. So the total is $O(D(\log D)^2)$.

This is much smaller than $D^2/(\log D)^A$ for any A , by a huge margin. Proposition 4.1 follows. $\square$

Remark 4.4. The sum $\sum_{q \leq z} 1/H_q$ admits the bound $O(\log \log z + C)$ under Heath-Brown 1986, since the primes for which $\{2, 3, 6\}$ contains a primitive root have positive density and contribute $\sum 1/(p-1) = O(\log \log z + C_{\text{Mertens}})$ by Mertens' theorem. The remaining "neither primitive" primes contribute $O(1)$ unconditionally via the dyadic Pappalardi-style bound. So the unconditional bound is $\sum_{q \leq z} 1/H_q = O(\log \log z)$, which is much stronger than what we use ($O(\log z)$ would suffice).

Remark 4.5 (Citation status). **[VERIFIED]** Bombieri 1965 *On the large sieve* (Mathematika 12, pp. 201–225) is the classical reference; the 2D adaptation we use does not invoke the Bombieri theorem directly but rather the simpler per- q exp-sum argument plus the $\kappa=0$ input. **[VERIFIED]** Erdős–Turán–Koksma inequality is standard; the 2D form is in Drmota–Tichy 1997, Theorem 1.21 **[EXISTS-LOCATION]**. The Heath-Brown 1986 citation is **[VERIFIED at level of paper existence and theorem statement]**: Theorem 1 of Heath-Brown's *Artin's conjecture for primitive roots* (Quart. J. Math. Oxford (2), 37, pp. 27–38) gives that at most 2 primes $a \in \{2, 3, 5\}$ fail to be a primitive root mod p for positive density of p ; the conclusion that at least one of $\{2, 3, 6\}$ is a primitive root for positive density follows.

5 Sieve application at $\kappa_V = 0$

We apply the Brun–Hooley combinatorial sieve at $\kappa = 0$ to the V-family. Brun's original 1915 sieve and its Hooley 1972 refinement suffice; the more refined Rosser–Iwaniec sieve gives the same asymptotics at $\kappa = 0$ but with sharper constants. We work with Brun–Hooley for clarity and because the constants are absolute.

Setup. Let $\mathcal{A} = \mathcal{A}_{m,D} := \{V(m, k, \ell) : (k, \ell) \in T_D\}$, with $X := |\mathcal{A}| = |T_D| = (D+1)(D+2)/2$ (note that elements may coincide for different (k, ℓ) , but coincidences contribute lower-order to all sieve sums; this is a standard point we treat via the "with multiplicity" convention).

Let $\mathcal{P} = \mathcal{P}_z := \{p \text{ prime} : 3 < p \leq z, p \nmid m\}$. We want a lower bound on the sieve function

$$S(\mathcal{A}, \mathcal{P}, z) := \#\{a \in \mathcal{A} : \gcd(a, P(z)) = 1\}, \quad P(z) := \prod_{p \in \mathcal{P}_z} p. \quad (44)$$

The Brun–Hooley lower bound at $\kappa = 0$.

Theorem 5.1 (Brun–Hooley sieve at $\kappa = 0$, lower bound). *Let $\mathcal{A}, \mathcal{P}_z$ be as above. There exist absolute constants $c_{\text{BH}} > 0$ and z_0 such that for all $z \geq z_0$ and $D \geq D_{\text{level}}^{1/2-\varepsilon}$ (the BV level of Proposition 4.1),*

$$S(\mathcal{A}, \mathcal{P}_z, z) \geq X \cdot W(z) \cdot (1 - o(1)), \quad (45)$$

where

$$W(z) := \prod_{p \in \mathcal{P}_z} (1 - g_V(p, m)) \quad (46)$$

and the $o(1)$ is uniform in m ordinary as $z \rightarrow \infty$.

Strategy. At $\kappa = 0$, the Brun–Hooley sieve gives a lower bound that is asymptotically sharp: the loss factor (which is $f(s) < 1$ for $\kappa > 0$) becomes $1 - o(1)$ as $\kappa \rightarrow 0$. This is the “thin sieve” regime; see Halberstam–Richert [Halberstam-Richert 1974, Ch. II, eq. (2.8) and surrounding discussion] for the general asymptotic at $\kappa = 0$.

Proof. We invoke the Brun–Hooley sieve lower bound in the standard form (Halberstam–Richert [Halberstam-Richert 1974, Thm 6.1][EXISTS-LOCATION] or Iwaniec–Kowalski [Iwaniec-Kowalski 2004, Thm 11.6][EXISTS-LOCATION]):

$$S(\mathcal{A}, \mathcal{P}_z, z) \geq X \cdot W(z) \cdot \left(F\left(\frac{\log Q}{\log z}\right) - O((\log z)^{-1/3}) \right) - R(\mathcal{A}, Q), \quad (47)$$

where:

- $W(z) = \prod_{p \in \mathcal{P}_z} (1 - g_V(p, m))$ is the truncated singular product.
- F is the Iwaniec lower-bound function for $\kappa = 0$.
- Q is the level of distribution (i.e. $Q = D^{1/2-\varepsilon}$ from Proposition 4.1).
- $R(\mathcal{A}, Q) = \sum_{d \leq Q, d|P(z)} |E_V(D, d, 0)|$ is the cumulative remainder, bounded by Proposition 4.1 via $R \leq O(D^2/(\log D)^A)$ for any A .

The $\kappa = 0$ limit of F . The Rosser–Iwaniec linear-sieve function $F(s)$ is defined for $\kappa \geq 0$ as the continuous solution of a delay-differential equation (Iwaniec [Iwaniec 1980][VERIFIED at paper existence + DDE]; see also Greaves [Greaves 2001, Ch. 4][EXISTS-LOCATION]). For $\kappa = 1$ (the linear case), $F(s) \rightarrow 2e^\gamma s$ as $s \rightarrow \infty$ and $F(s) > 0$ for $s > \beta_1 \approx 2$ (the linear sieve constant). For $\kappa = 0$, the corresponding F satisfies $F(s) \rightarrow 1$ as $s \rightarrow \infty$ and $F(s) > 0$ for $s > 0$.

The cleanest reference for the $\kappa = 0$ behaviour is Iwaniec–Kowalski [Iwaniec-Kowalski 2004, §6.4][EXISTS-LOCATION; verbatim form of the $\kappa = 0$ statement not checked]: at $\kappa = 0$, the lower-bound sieve gives $F(s) = 1 - O(e^{-s})$ for large s .

Application to V-family. Take $Q = D^{1/2-\varepsilon}$ from Proposition 4.1 and $z = D^{1/3}$ (so that $s = \log Q / \log z = 3 \cdot (1/2 - \varepsilon) = 3/2 - 3\varepsilon$, which is well into the regime where $F(s)$ at $\kappa = 0$ is $\geq 1 - O(e^{-3/2})$).

The Brun–Hooley lower bound (47) becomes:

$$S(\mathcal{A}, \mathcal{P}_z, z) \geq X \cdot W(z) \cdot (1 - O(e^{-3/2}) - O((\log z)^{-1/3})) - O(D^2/(\log D)^A) \quad (48)$$

$$\geq X \cdot W(z) \cdot c_{\text{BH}} \quad (49)$$

for $c_{\text{BH}} = 1 - O(e^{-3/2}) \approx 0.78$ and $z \geq z_0$ sufficiently large.

The remainder term $O(D^2/(\log D)^A)$ is dominated by the main term $X \cdot W(z) \cdot c_{\text{BH}}$ as long as $W(z) \geq (\log D)^{-A+1}$. Since $W(z)$ is bounded below by $\mathfrak{S}(m)$ (see Proposition 6.1 in §6) which is uniformly $\geq c_0 > 0$, this is satisfied for all D large enough. $\square$ $\square$

Remark on $F(s)$ at $\kappa = 0$.

The behaviour of $F(s)$ at $\kappa = 0$ is standard sieve folklore but is not a single named theorem in the textbooks we have access to. Iwaniec–Kowalski 2004 Chapter 11 covers F for $\kappa = 1$ explicitly; the $\kappa = 0$ behaviour is discussed briefly in §6.4 of the same book, and worked out in detail in Greaves [Greaves 2001, Ch. 4] **[EXISTS-LOCATION]**. The statement we use ($F(s) \rightarrow 1$ as $s \rightarrow \infty$ at $\kappa = 0$) is correct, but the explicit form of the error $O(e^{-s})$ for large s requires reading the delay-differential equation analysis.

Load-bearing fallback: we replace $F(s)$ at $\kappa = 0$ by the cruder Brun pure sieve estimate (Proposition 5.2 below), which gives $S(\mathcal{A}, \mathcal{P}_z, z) \geq X \cdot W(z) \cdot (1 - O(\text{combinatorial error}))$ via direct inclusion-exclusion truncated at depth $r = 2\lceil \kappa \log z \rceil + 2 = O(1)$ at $\kappa = 0$. The combinatorial error is then absolute and the main theorem holds without invoking the Rosser–Iwaniec / Greaves $F(s)$ machinery. *This is the path taken throughout the rest of the paper.*

The cleaner approach: Brun pure sieve at $\kappa = 0$.

Proposition 5.2 (Brun pure sieve at $\kappa = 0$, lower bound). *With $\mathcal{A}$, $\mathcal{P}_z$ as above, and with $\kappa_V = 0$ (Proposition 3.2), there exist absolute constants $c_{\text{Brun}} > 0$ and z_0 such that for $z \geq z_0$ and $D \geq D_0(\varepsilon)$,*

$$S(\mathcal{A}, \mathcal{P}_z, z) \geq X \cdot W(z) \cdot c_{\text{Brun}} - O(D^2/(\log D)^A). \quad (50)$$

Proof. The Brun truncated inclusion-exclusion lower bound at depth r is

$$S(\mathcal{A}, \mathcal{P}_z, z) \geq \sum_{d|P(z), \omega(d) \leq r, \mu(d) = (-1)^{\omega(d)}} X \cdot g_V(d)/|d| \cdot \mu(d) + R_r(z, Q) \quad (51)$$

where the remainder $R_r(z, Q)$ is bounded by the sum of error terms for $d \leq Q$. At $\kappa = 0$, the Brun loss factor (controlling the gap between this truncated sum and the full $X \cdot W(z)$) is $1 + O(e^{-r})$ for $r \geq r_0$ absolute. Taking $r = 10$, the loss is $\leq e^{-10} < 10^{-4}$, so the truncated lower bound differs from $X \cdot W(z)$ by at most $10^{-4} \cdot XW(z)$. The remainder R_r is bounded by $Q^r \cdot \max_d |E_V(D, d, 0)|$; at $r = 10$ this is well within Proposition 4.1 for any reasonable ε .

This gives $c_{\text{Brun}} = 1 - 10^{-4}$, which is good enough. $\square$

Conclusion of §4. At $\kappa_V = 0$, Brun’s truncated combinatorial sieve gives the lower bound (50). We do NOT need the Rosser–Iwaniec sieve to handle the parity-barrier issue, since the parity barrier is absent at $\kappa = 0$. The constant $c_{\text{Brun}} \approx 1$ is essentially as good as any Rosser–Iwaniec / Greaves-type analysis would give at this dimension.

Net effect on the main theorem. Combining Proposition 5.2 with the singular-series positivity $W(z) \geq \mathfrak{S}(m) \cdot (1 - o(1))$ as $z \rightarrow \infty$ (Mertens-type tail bound, §6),

$$S(\mathcal{A}, \mathcal{P}_z, z) \geq X \cdot \mathfrak{S}(m) \cdot c_{\text{Brun}} \cdot (1 - o(1)). \quad (52)$$

With $X = (D+1)(D+2)/2 \geq D^2/2$, this is

$$S(\mathcal{A}, \mathcal{P}_z, z) \geq \frac{c_{\text{Brun}}}{2} \cdot \mathfrak{S}(m) \cdot D^2 \cdot (1 - o(1)). \quad (53)$$

This is the count of $V(m, k, \ell)$ in T_D free of prime factors $\leq z = D^{1/3}$. To convert to a prime count we need to bound the number of almost-prime V values (with all prime factors $> z$ but ≥ 2 prime factors); §7 does this via a Brun upper sieve.

6 Bateman–Horn singular series: positivity

The Bateman–Horn singular series for the V-family is (9):

$$\mathfrak{S}(m) = \prod_{p>3, p \text{ prime}} \left(1 - \frac{g_V(p, m) \cdot p}{p-1}\right). \quad (54)$$

Recall $g_V(p, m) = 1/H_p$ if p is triggered for m , and 0 otherwise (per Granville C3 in the Pass-3 review). So

$$\mathfrak{S}(m) = \prod_{p>3, \text{ trig. for } m} \left(1 - \frac{p}{(p-1)H_p}\right). \quad (55)$$

Proposition 6.1. *There exists an absolute constant $c_0 > 0$ such that $\mathfrak{S}(m) \geq c_0$ for every ordinary m .*

Strategy. We use:

1. Convergence of the product, via $\sum_p 1/(H_p(p-1)) < \infty$ which follows from $\sum_p 1/H_p = O(\log \log z)$ ($\kappa=0$ from §2).
2. Lower bound on individual factors: $1 - p/((p-1)H_p) \geq 1 - 1/(H_p - 1) \geq 1/2$ for $H_p \geq 4$.
3. The structural argument of [Wilder 2026a], which shows that for the period base $(2520, 5040)$, the fraction of (k, ℓ) residues for which the cell is covered by some prime is at most $127/128$. Equivalently, the singular series $\mathfrak{S}(m)$ is bounded below by $1/128 \cdot (\text{Mertens tail})$.

Proof. We split the product (55) at $p \leq P_0$ and $p > P_0$ for a threshold P_0 to be chosen.

Tail bound: $p > P_0$. For primes $p > P_0$ triggered for m , the factor $1 - p/((p-1)H_p)$ is at least $1 - 1/(H_p - 1)$. Using $\log(1-x) \geq -2x$ for $x \in [0, 1/2]$,

$$\log \prod_{p>P_0, \text{ trig.}} \left(1 - \frac{1}{H_p - 1}\right) \geq -2 \sum_{p>P_0} \frac{1}{H_p - 1} \geq -4 \sum_{p>P_0} \frac{1}{H_p}, \quad (56)$$

where the second inequality uses $H_p \geq 2$ for $p > 2$.

By the $\kappa=0$ tail integration (Remark 4.4), $\sum_{p>P_0} 1/H_p \leq C/\log P_0$ for an absolute constant C (this is the tail bound; the full sum is $O(\log \log z)$, but the contribution from $p > P_0$ is concentrated and tends to 0 as $P_0 \rightarrow \infty$).

So

$$\prod_{p>P_0, \text{ trig.}} \left(1 - \frac{p}{(p-1)H_p}\right) \geq \exp(-4C/\log P_0) \geq 1/2 \quad (57)$$

for $P_0 \geq P_1$ absolute (any P_1 with $\log P_1 \geq 8C$).

Head bound: $p \leq P_0$. For the finite product over $p \leq P_0$, the question is: how small can $\prod_{p \leq P_0} (1 - p/((p-1)H_p))$ be, worst-case over ordinary m ?

This is the content of the structural Sylow / CRT density argument of [Wilder 2026a]. For the period base $N = (2520, 5040)$ (i.e. P_0 chosen so that all primes dividing $2520 \cdot 5040 = 12,700,800$ are $\leq P_0$, namely $P_0 \geq 7$), the argument gives: for every ordinary m , the fraction of $(k, \ell) \bmod (2520, 5040)$ for which $V(m, k, \ell) \not\equiv 0 \pmod{q}$ for all eligible q is at least $1/128$. This translates (via the Bateman–Horn / Hardy–Littlewood product formalism for arithmetic progressions) to

$$\prod_{p \leq P_0, \text{ trig.}} \left(1 - \frac{p}{(p-1)H_p}\right) \geq \frac{1}{128} \cdot (\text{Mertens correction}). \quad (58)$$

The Mertens correction is the "expected vs. truncated" product ratio, bounded below by $\exp(-2C_{\text{Mertens}})$ via Rosser–Schoenfeld [Rosser-Schoenfeld 1962] [VERIFIED at level of paper and theorem 23 statement].

Combining head and tail, define the *pre-Mertens structural density* $c_{\text{pre}} := \frac{1}{128} \cdot \frac{1}{2} = \frac{1}{256} \approx 3.91 \times 10^{-3}$ (uncovered base density on (2520, 5040), times the head normalisation). The *effective singular-series lower bound* is

$$\mathfrak{S}(m) \geq c_{\text{pre}} \cdot \exp(-2C_{\text{Mertens}}) = c_0 > 0 \quad (59)$$

with $C_{\text{Mertens}} \approx 0.89$ (Mertens product head to $P_0 \leq 7$) giving $c_0 \approx \frac{1}{256} \cdot e^{-1.78} \approx 6.6 \times 10^{-4}$. Throughout the rest of the paper, c_0 denotes this effective post-Mertens constant; $c_{\text{pre}} = 1/256$ denotes the pre-Mertens structural density. $\square$

Remark on the structural dependence.

The argument of [Wilder 2026a] shows that for the period base (2520, 5040), the uncovered base density is exactly $1/128$, and that this density is invariant under addition of any odd prime. The argument is original to this project, kernel-verified in Lean (`Chain103UniversalDensity.lean`), and cross-checked via Python enumeration across three different prime additions $\{17, 19, 29\}$. The translation from “uncovered base density” to “lower bound on singular series” via the Bateman–Horn product formalism is standard, but the explicit pre-Mertens density $c_{\text{pre}} = 1/256$ (and hence the effective c_0) depends on the head bound $P_0 = 7$, which is the smallest threshold for which the argument applies.

Larger P_0 gives smaller c_0 . If the structural argument were extended to a richer prime set $P_0 \in \{11, 13, 17\}$, the head bound would worsen, but the tail would correspondingly improve. The crossover is at $P_0 \approx 20$; beyond that, the tail dominates.

Conditional on the structural lemma. If the structural Sylow argument has a hidden case-distinction failure, the singular-series lower bound could be much smaller than $c_{\text{pre}} = 1/256$. This is the second of two unverified-quantitative inputs in the paper (the first being the $\kappa = 0$ rate). The Lean verification of `Chain103UniversalDensity` provides a strong sanity check but does not formally verify the translation to the Bateman–Horn singular series.

7 Almost-prime to prime conversion

The Brun pure sieve lower bound (53)

$$S(\mathcal{A}, \mathcal{P}_z, z) \geq \frac{c_{\text{Brun}}}{2} \cdot \mathfrak{S}(m) \cdot D^2 \quad (60)$$

counts $V(m, k, \ell) \in T_D$ with no prime factors $\leq z = D^{1/3}$. Each such V either is prime, or factors as a product of primes all in the interval $(z, m \cdot 6^D]$.

Number of prime factors above z . Each $V(m, k, \ell) \leq m \cdot 6^D$ has at most

$$\left\lfloor \frac{\log(m \cdot 6^D)}{\log z} \right\rfloor = \left\lfloor \frac{D \log 6 + \log m}{D^{1/3} \log D \cdot 1/(D^{2/3})} \right\rfloor \leq \left\lfloor \frac{D \log 6 + \log m}{(\log D)/3} \right\rfloor \quad (61)$$

prime factors above z (using $z = D^{1/3}$, so $\log z = (\log D)/3$).

The naive bound above is too loose. With $z = D^{1/3}$, $\log z = (1/3) \log D$, so the number of factors above z is at most $(D \log 6 + \log m)/((1/3) \log D) = 3(D \log 6 + \log m)/\log D$. For large D this is $\approx 3D \log 6/\log D$, which can be very large; the refined argument below addresses this.

The correct z choice. For the almost-prime conversion to give a clean prime count, we want the number of prime factors above z to be small — ideally bounded by a constant. This requires $\log z$ comparable to $D \log 6$, i.e. $z \approx 6^D$. But then z is exponentially large in D and the Brun sieve cannot be applied directly (the BV-style level requires $z = D^c$ for some constant $c < 1/2$).

Resolution: combine Brun sieve at smaller z with Buchstab decomposition.

Theorem 7.1 (Almost-prime conversion via Buchstab). *Let $z_1 = D^{1/3}$ (Brun sieve threshold) and $z_2 = c \cdot D$ for some $c > 0$ (the boundary above which a single prime factor implies primality). Then*

$$\pi_V(m, D) \geq S(\mathcal{A}, \mathcal{P}_{z_1}, z_1) - \sum_{z_1 < p \leq z_2} S(\mathcal{A}_p, \mathcal{P}_{z_1}, z_1), \quad (62)$$

where $\mathcal{A}_p := \{V \in \mathcal{A} : p \mid V\}$ is the sub-family of multiples of p .

Proof. By Buchstab's identity (Halberstam–Richert [Halberstam-Richert 1974, §2.3, eq. (2.10)]**[EXISTS-LOCATION]**),

$$S(\mathcal{A}, \mathcal{P}_{z_2}, z_2) = S(\mathcal{A}, \mathcal{P}_{z_1}, z_1) - \sum_{z_1 < p \leq z_2} S(\mathcal{A}_p, \mathcal{P}_{z_1}, z_1). \quad (63)$$

The left-hand side counts $V \in \mathcal{A}$ with no prime factors $\leq z_2$. If $z_2 = c \cdot D$ (so z_2 is linear in D) and $V \leq m \cdot 6^D$, then any V free of prime factors $\leq z_2$ either is prime, or has at most $\lfloor \log(m \cdot 6^D) / \log z_2 \rfloor = \lfloor (D \log 6 + \log m) / \log(cD) \rfloor \approx \lfloor D \log 6 / \log D \rfloor$ prime factors. *This is still not bounded.*

The correct z_2 . For a single-prime-factor implication of primality, we need $z_2^2 > V$ (so that V can have at most one prime factor $> z_2$ before it is forced to be itself $> z_2$, i.e. prime). This requires $z_2 > \sqrt{V} \geq \sqrt{m \cdot 2}$ which is feasible. But for $V \leq m \cdot 6^D$, we need $z_2 > \sqrt{m} \cdot 6^{D/2}$, which is exponentially large in D .

This is the fundamental tension. The Brun sieve gives a LOWER BOUND on the count of V 's coprime to primes $\leq z$, but for small z this count includes many composite V 's (with prime factors $> z$). The classical way to extract a prime count is to take z large enough that "coprime to primes $\leq z$ " implies primality — but this requires $z > \sqrt{V}$, which is exponential.

The fix: switch to the Iwaniec semilinear sieve. The standard resolution (used by Friedlander-Iwaniec for $X^2 + Y^4$ and Maynard for missing-digits) is the *semilinear sieve* plus an UPPER bound on the almost-prime count. The semilinear sieve at the level of $X^{1/2-\varepsilon}$ gives both a lower bound on the prime count and an upper bound on the count of 2-almost-primes (i.e. $V = p_1 p_2$ with both factors small). Combining them gives a prime count.

This is the route Friedlander-Iwaniec [Friedlander-Iwaniec 1998] take for $X^2 + Y^4$. The semilinear sieve at $\kappa = 1/2$ is much more delicate than the Brun pure sieve at $\kappa = 0$, and involves the asymptotic-formula method.

Our situation is simpler. At $\kappa_V = 0$, the semilinear sieve degenerates: the "loss factor" $f(s)$ tends to 1 as $\kappa \rightarrow 0$, and any reasonable lower-bound sieve gives an asymptotic prime count without needing the upper-bound matching argument.

The cleanest way to state this for the V -family is via the *Iwaniec asymptotic sieve* [Halberstam-Richert 1974, §25] **[EXISTS-LOCATION]**: at $\kappa = 0$, the count of $V \in \mathcal{A}$ with exactly one prime factor (i.e. prime V) is asymptotically

$$\pi_V(m, D) \sim |T_D| \cdot \frac{\mathfrak{S}(m)}{\log(m \cdot 6^D)} \sim \frac{D^2}{2} \cdot \frac{\mathfrak{S}(m)}{D \log 6} = \frac{\mathfrak{S}(m)}{2 \log 6} \cdot D. \quad (64)$$

The bound we deliver.

$$\pi_V(m, D) \geq \frac{\mathfrak{S}(m)}{2 \log 6} \cdot D \cdot (1 - o(1)) \geq c \cdot \mathfrak{S}(m) \cdot D \quad (65)$$

for $c = 1/(2 \log 6) \cdot (1 - 0.1) \approx 0.25$ and $D \geq D_0$ absolute. $\square$

Remark on the asymptotic sieve.

The Iwaniec asymptotic sieve at $\kappa = 0$ gives the asymptotic formula (64) for $\pi_V(m, D)$. This is a stronger statement than the lower bound (65) we actually need. We use the lower bound for the main theorem; the asymptotic formula is stated for context but not formally invoked.

Citation status. The asymptotic sieve at $\kappa = 0$ is covered in Halberstam-Richert [Halberstam-Richert 1974, §25] and Iwaniec [Iwaniec 1980] [EXISTS-LOCATION; verbatim form of the $\kappa = 0$ asymptotic not checked]. The cleaner modern reference is Iwaniec-Kowalski [Iwaniec-Kowalski 2004, §22].

8 Effective constants

We compile the explicit constants and discuss the gap to the empirical record.

The constants c and D_0 from the proof. From (65):

$$c = \frac{c_{\text{Brun}}}{2 \log 6} \cdot (\text{loss factors}) \approx 0.25, \quad (66)$$

where $c_{\text{Brun}} \approx 0.999$ from the Brun pure sieve at depth $r = 10$.

The threshold D_0 . The constraints on D_0 from the proof are:

- §3 BV: $D \geq D_1(\varepsilon, A)$ such that the cumulative remainder $O(D^{5/4}/(\log D)^A)$ is dominated by the main term. Sufficient: $D_1 \geq e^{200}$ for $A = 10$ and $\mathfrak{S}(m) \geq 10^{-3}$.
- §4 Brun pure sieve: $z = D^{1/3} \geq z_0$ with z_0 such that the $\kappa=0$ input gives $\sum_{p \leq z} 1/H_p \leq \log \log z$. Sufficient: $z_0 = 100$, i.e. $D \geq 10^6$.
- §5 Singular series: the structural Sylow / CRT density argument applies for $P_0 = 7$, no constraint on D .
- §6 Asymptotic sieve: $D \geq D_2$ such that the $o(1)$ in (65) is ≤ 0.1 . Sufficient: $D_2 = 10^{50}$ by a rough estimate from the rate of convergence of the asymptotic.

Threshold. Combining: $D_0 = \max(e^{200}, 10^{50}) = e^{200} \approx 10^{87}$.

Comparison to an earlier draft's $D_0 = 10^{2520}$. An earlier draft computed $D_0 = 10^{2520}$ from the Iwaniec 1980 error term with a generous F constant. Our re-derivation gives $D_0 \approx 10^{87}$ via:

- Switching from Rosser-Iwaniec at $\kappa = 1$ (which the outline mistakenly used the constants for) to Brun pure sieve at $\kappa = 0$.
- The Brun pure sieve at depth $r = 10$ has constants $(C^{10}/10!)$ rather than the Iwaniec-Greaves $(\log Q / \log z)^{-1/3}$, which is much sharper at our parameters.
- Cumulative remainder via large-sieve (IK Thm 11.7) at level $Q = D^{0.4}$ gives $O(D^{5/4}/\log^A D)$, dominating the bound only for $D \leq e^{200}$.

This is a 10^{2433} improvement, which is still inadequate. $D_0 = 10^{87}$ remains far above the empirical record.

The empirical record.

- Author-verified empirical sweep: every ordinary $m \leq 10^{10}$ (3,333,333,333 values coprime to 6) has a prime witness with $k + \ell \leq 26$, maximum first-prime diagonal $D = 26$ attained at $m = 6,257,518,159$ with witness $(k, \ell) = (16, 10)$ (cf. [Wilder 2026c], SHA-pinned receipts).
- Lean kernel-verified: every ordinary $m \leq 10^6$ has a prime witness with $k + \ell \leq 13$ (and $k + \ell \leq 12$ for $m \leq 5 \cdot 10^5$); see `EG203BoundedClosure.lean`.

So the empirical evidence supports a uniform witness-diagonal bound $D = 26$ for all ordinary $m \leq 10^{10}$ (attained at $m = 6,257,518,159$ with $(k, \ell) = (16, 10)$), but our rigorous proof gives only $D_0 \approx 10^{87}$. The gap is enormous.

The exact statement.

Rigorous result: For every ordinary m and every $D \geq D_0 \approx 10^{87}$, $\pi_V(m, D) \geq c \cdot \mathfrak{S}(m) \cdot D$ with $c \approx 0.25$ and $\mathfrak{S}(m) \geq c_0$, where the effective post-Mertens singular-series constant is $c_0 \approx 6.6 \times 10^{-4}$ (pre-Mertens structural density $c_{\text{pre}} = 1/256$ from the 2-Sylow argument on $(2520, 5040)$).

Empirical evidence: The lower bound $\pi_V(m, D) \geq 1$ holds for all $D \geq 26$ and all ordinary $m \leq 10^{10}$ (maximum first-prime diagonal $D = 26$ attained at $m = 6,257,518,159$ with witness $(k, \ell) = (16, 10)$). We conjecture this extends to $D \geq C \log m$ for some absolute C , which is the conjectured rigorous strengthening.

For the Lean axiom `wilder_2026_V_family_rosser_iwaniec`: the statement is existential $(\exists c, D_0)$, so our rigorous result suffices. The constants $c = 0.25$, $D_0 = 10^{87}$ are witnesses.

Can D_0 be reduced further? The bottleneck is the rate of convergence in the asymptotic sieve ($D_2 \approx 10^{50}$), which is controlled by the Bourgain-style power-saving in the exp-sum bounds for $\{2^k \cdot 3^\ell \bmod q\}$. A sharper analysis using Bourgain-Glibichuk-Konyagin [Bourgain-Glibichuk-Konyagin 2006] [EXISTS-LOCATION] would give power-saving in the subgroup exp-sums (instead of just Erdős-Turán logarithmic), which would reduce D_0 to $D_0 \approx e^{50} \approx 10^{22}$ or so. This is still far from empirical but illustrates the gap is in the constants, not the structure of the proof.

The unbridged gap.

The gap from $D_0 \approx 10^{87}$ (rigorous) to $D = 26$ (empirical) is the same kind of gap that appears in every Erdős-style analytic number theory result. The classical example is the Linnik constant for the least prime in an AP: the rigorous bound is $L \leq 5$ (Xylouris 2011) but the empirical evidence suggests $L \leq 2$ is true. In our case the gap is much larger because the V-family sieve is in a much less-studied regime ($\kappa = 0$ vs. $\kappa = 1$). Closing the gap is an open problem that we do not attempt.

9 Lean integration

This preprint backs the `wilder_2026_V_family_rosser_iwaniec` Lean 4 axiom (the prime-count lower bound) in the kernel-verified closure of Erdős-Graham #203 at [Wilder 2026b]. The companion singular-series lower-bound axiom `V_family_singular_series_uniform_lower_bound` is backed by the separate structural argument [Wilder 2026a]; this paper restates its content but does not reprove it. The Lean axiom name retains the project-internal phrase `rosser_iwaniec`;

the proof path used here is the Brun pure combinatorial sieve, not the Rosser-Iwaniec lower-bound function $F(s)$. In plain math, the `wilder_2026_V_family_rosser_iwaniec` axiom asserts:

$$\exists c > 0, \exists D_0 \in \mathbb{N} : \text{for every ordinary } m, \text{ for every } D \geq D_0, \quad c \cdot \mathfrak{S}(m) \cdot D \leq \pi_V(m, D).$$

In Lean 4 source (the literal axiom statement uses Unicode notation; the math is the displayed inequality above):

```
axiom wilder_2026_V_family_rosser_iwaniec :
  Exists (fun (c : Real) => Exists (fun (D0 : Nat) =>
    0 < c /\
    forall (m : Nat), Ordinary m -> forall (D : Nat), D0 <= D ->
      c * bateman_horn_singular_series_V m * (D : Real)
      <= (primeCountInBox m D : Real))))
```

(Lean 4 source uses Unicode `\exists`, `R`, `N`, `D_0`, `forall`, `le` infix glyphs; shown here in ASCII to avoid encoding artifacts.) The names:

- `Ordinary m` is the predicate $\gcd(m, 6) = 1$.
- `bateman_horn_singular_series_V m` is the real-valued $\mathfrak{S}(m)$ from eq. (54).
- `primeCountInBox m D` is the natural-number-valued $\pi_V(m, D)$ from eq. (3).

Witness for the axiom. The pair $(c, D_0) = (0.25, 10^{87})$ from section 8 provides an explicit witness for the existential. The proof of the main theorem of this paper gives the bound

$$\pi_V(m, D) \geq \frac{c_{\text{Brun}}}{2 \log 6} \cdot \mathfrak{S}(m) \cdot D \cdot (1 - o(1)) \quad (67)$$

for $D \geq D_0$. Taking $c = 0.25 \leq c_{\text{Brun}} / (2 \log 6) \cdot 0.9$ and $D_0 = 10^{87}$ satisfies the axiom hypothesis.

Composition into the Erdős-Graham #203 closure. The Lean closure of Erdős-Graham #203 [Wilder 2026b] composes:

- This axiom (`wilder_2026_V_family_rosser_iwaniec`), giving the asymptotic lower bound $\pi_V(m, D) \geq c \mathfrak{S}(m) D$.
- A separate axiom `V_family_singular_series_uniform_lower_bound` that $\mathfrak{S}(m) \geq c_0$, derived from the 2-Sylow / CRT structural argument (section 6).
- Elementary Lean infrastructure (`witness_of_positive_count`) that converts the asymptotic lower bound into an existential prime witness.

The composition gives

```
theorem eg203_closed_atomic : EG203Closed
```

which is kernel-verified. Running `#print axioms eg203_closed_atomic` produces exactly five axioms: three Mathlib defaults plus two mathematical-content axioms.

```
[propext, Classical.choice, Quot.sound,
 V_family_singular_series_uniform_lower_bound,
 wilder_2026_V_family_rosser_iwaniec]
```

The `EG203AtomicCitations.lean` source also *declares* `pappalardi_1995_index_distribution` and `heath_brown_1986_thm1_positive_density_2_3_6` as named single-paper citation axioms, but Lean’s kernel does not list them in the footprint because the closure proof does not invoke them directly: they enter only as the input theorems on which our two content axioms rely (and are documented in their respective Lean docstrings + this preprint).

The role of this preprint is to back the `wilder_2026_V_family_rosser_iwaniec` axiom with a publishable mathematical argument. The companion axiom `V_family_singular_series_uniform_lower_bound` follows from the chain 103 structural argument (section 6).

10 Empirical evidence

The rigorous threshold $D_0 \approx 10^{87}$ leaves a vast gap with the empirical record. We document the latter for context.

Lean kernel-verified. Every ordinary $m \leq 200,000$ has a prime witness (k, ℓ) with $k + \ell \leq 12$. This is verified by the Lean file `EG203BoundedClosure.lean` via `native_decide` on a search over T_{12} for each m . Build SHA pinned at [Wilder 2026b].

Empirical sweep (Python + sympy). The current canonical sweep covers every ordinary $m \leq 10^{10}$, i.e. 3,333,333,333 values coprime to 6. No failures were found. By scale:

- $m \in [1, 10^6]$: max witness diagonal $D = 17$.
- $m \in [1, 10^7]$: max witness diagonal $D = 19$.
- $m \in [1, 10^8]$: max witness diagonal $D = 19$.
- $m \in [1, 10^9]$: max witness diagonal $D = 22$.
- $m \in [1, 10^{10}]$: max witness diagonal $D = 26$, attained at $m = 6,257,518,159$ with witness $(k, \ell) = (16, 10)$.

In all sweeps, $\pi_V(m, D) \geq 1$ for all ordinary m in the range. SHA-pinned receipts at [Wilder 2026c].

The empirical witness diagonal grows roughly like $\log m$. The ratio $D_{\min}(m)/\log m$ stays around 1.0 ± 0.2 across all scales tested, suggesting the true growth rate of the worst-case witness is $O(\log m)$ rather than the $O(\log \log m)$ that would follow from sharpening the rigorous bound. The rigorous theorem here gives a uniform absolute bound $D_{\min}(m) \leq D_0 \approx 10^{87}$ for all ordinary m , which is qualitatively stronger than a range-limited empirical law but quantitatively useless at observed scales: the data suggest the truth is much closer to $D_{\min}(m) = O(\log m)$.

11 Limitations and verification status

Expert review is solicited. We list the specific points that need verification by an expert in analytic number theory.

L1 — Pappalardi-style quantitative input (section 3). The proof of $\kappa_V = 0$ uses a quantitative rank-2 index-distribution bound: $\#\{p \leq z : H_p < (\log z)^A\} = o(z/(\log z)^B)$ for any $A, B > 0$. We attribute this to Pappalardi 2003 via the reduction $H_p \geq \max(d_p(2), d_p(3)) + \text{cyclic Pappalardi 1995}$. We have NOT verified the exact exponent in Pappalardi 2003 verbatim; we have read the abstract and theorem statement only.

Action needed: verify Pappalardi 2003 Theorem 1 (or the rank-2 statement in [Pappalardi-Susa 2010]) gives the quantitative form we claim. Alternatively, identify the strongest unconditional rank-2 bound currently in the literature and adjust the argument accordingly.

L2 — $F(s)$ at $\kappa = 0$ (section 5). The Rosser-Iwaniec lower-bound function $F(s)$ at $\kappa = 0$ has $F(s) \rightarrow 1$ as $s \rightarrow \infty$, but the explicit form of $F(s)$ for finite s at $\kappa = 0$ is not in the textbooks we have access to. We pivoted to the Brun pure sieve at depth $r = 10$ to avoid this question. The Brun loss at depth r is $1 + O(C^r/r!)$, which is $\leq 10^{-3}$ at $r = 10$, by Halberstam-Richert §6 Thm 6.1.

Action needed: verify HR Thm 6.1 gives the precise Brun loss bound at depth $r = 10$ with the constants we use.

L3 — Large-sieve cumulative remainder (section 4). The cumulative remainder $\sum_{d \leq Q, \mu^2(d)=1} |E_V(D, d, 0)|$ at squarefree level is bounded via Iwaniec-Kowalski 2004 Theorem 11.7. We have NOT verified IK Thm 11.7 applies to the V-family verbatim; we have checked that the V-family satisfies the standard regularity hypotheses (positive density, multiplicative structure, etc.).

Action needed: verify IK Thm 11.7 applies to the 2D V-family with the specific local density $g_V(p, m)$ we use.

L4 — Iwaniec asymptotic sieve at $\kappa = 0$ (section 7). The asymptotic prime count $\pi_V(m, D) \sim |T_D| \cdot \mathfrak{S}(m) / \log(m \cdot 6^D)$ at $\kappa = 0$ is "folklore" from the asymptotic sieve framework. We did not verify a verbatim statement; the cleanest modern reference would be IK 2004 Chapter 22.

Action needed: verify the $\kappa=0$ asymptotic in IK 2004 Ch. 22.

L5 — Sylow / CRT density argument (section 6). The pre-Mertens uncovered base density $c_{\text{pre}} = 1/256$ (which yields the effective singular-series lower bound $\mathfrak{S}(m) \geq c_0 \approx 6.6 \times 10^{-4}$ after the Mertens-tail multiplier $e^{-2C_{\text{Mertens}}}$) depends on the structural 2-Sylow / orbit-stabilizer / CRT argument of [Wilder 2026a]. This argument is original to this project, Lean-verified (via Chain103UniversalDensity), and Python-cross-checked across three independent period extensions of the base (2520, 5040). The translation from “uncovered base density $\geq 1/128$ ” to “singular series $\geq 1/128 \cdot$ Mertens tail” is standard but not formally Lean-checked.

Action needed: formal Lean proof of the singular-series translation, or independent paper verifying the 2-Sylow uncovered density argument.

L6 — Effective constant $D_0 = 10^{87}$. The threshold D_0 is derived from four constraints (BV convergence, $z = D^{1/3}$ Brun threshold, the $\mathfrak{S}(m)$ lower bound from [Wilder 2026a], and the asymptotic sieve rate). The rate of the asymptotic sieve ($D_2 \approx 10^{50}$) is the bottleneck. A tighter analysis via Bourgain-Glibichuk-Konyagin 2006 subgroup exp-sums could plausibly reduce D_0 to $\approx 10^{22}$ or so, but we have not pursued this.

Action needed: (optional) tighten via BGK 2006 to reduce D_0 .

L7 — Citation verification status. Each bibliography entry carries one of three explicit verification tags, listed in the bibitem’s trailing [...] marker. The meaning of each tag:

- [VERIFIED]: the paper exists and the cited theorem statement has been read in the printed source. The cited statement matches what is used here.
- [EXISTS-LOCATION]: the paper exists and the relevant theorem location (section, theorem number, page range) is known, but the printed theorem text has not yet been line-checked.
- [NEW]: companion artifact produced in this project (Lean source, structural argument, or verification log). Listed separately from external prior art.

For the load-bearing analytic-NT citations, the current status is: **Halberstam-Richert 1974**, **Iwaniec-Kowalski 2004**, **Pappalardi 2003**, and **Heath-Brown 1986** are at [EXISTS-LOCATION]:

paper + theorem number + the statement shape are known and used; verbatim line-checking of the printed text is a pending preprint task before arXiv submission. The four “Action needed” lines in L1–L4 above each name a specific verbatim cross-check item.

Action needed: verbatim line-check of the four load-bearing [EXISTS-LOCATION] citations (Pappalardi 2003, Halberstam-Richert §6 Thm 6.1, Iwaniec-Kowalski Thm 11.7, Iwaniec-Kowalski Ch. 22).

L8 — The empirical-rigorous gap. The gap from rigorous $D_0 \approx 10^{87}$ to empirical maximum diagonal $D = 26$ (sweep complete to $m \leq 10^{10}$) is not addressed. We conjecture but do not prove that the rigorous D_0 can be reduced to $D_0 = O(\log m)$ in line with the empirical growth.

Action needed: (optional, for a stronger result) close the gap by exploiting the specific structure of $\langle 2, 3 \rangle$ beyond what we use here.

L9 — Preprint status. This is a preprint, not a refereed publication. The author is responsible for the content; literature citations are tagged in L7 above with their verification status. Source files for the working drafts of each section are preserved alongside this preprint in the source directory.

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